



12. If an update to a persistent array must be crash-safe, which method should be used?
 A) JCSysm.arrayCopyNonAtomic() B) JCSysm.arrayFillNonAtomic()
 C) JCSysm D) JCSysm.arrayCopy()
13. Payment card type approval testing is primarily based on which standard?
 A) ISO/IEC 14443 B) ISO/IEC 7816 and EMV specifications
 C) PCI DSS D) IEEE 802.11
14. Which status word represents successful execution in ISO7816?
 A) 0x9000 B) 0x6D00
 C) 0x1231 D) 0x6F32
15. In the response APDU (as per ISO 7816), what do SW1 and SW2 represent?
 A) Command header values B) Response status words
 C) Status words indicating execution result D) Command parameters
16. What is the main purpose of the personalization phase of payment cards?
 A) Testing the durability of plastic B) Assigning PCI certification to the card issuer
 C) Printing holograms and logos D) Encoding cardholder data, keys, and applications onto the card
17. What is the main purpose of Global Platform Card Architecture?
 A) To provide a secure framework for application management on smart cards B) To certify POS terminals
 C) To manufacture chip hardware D) To define cardholder banking procedures
18. What is the purpose of logical channels in Global Platform cards?
 A) To increase APDU buffer size B) To allow multiple applications to be selected and communicated with simultaneously
 C) To decrease APDU buffer size D) To encrypt card communication
19. What is the main purpose of key management in Global Platform?
 A) To store cardholder PINs B) To increase card memory space
 C) To configure terminal hardware D) To manage secure channels, authentication, and application loading securely
20. Which command is used to remove applications, packages, or security domains from a card?
 A) DELETE B) LOCK
 C) INSTALL D) LOAD

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the various types of smart cards and their usage with necessary example.	08	L*2	1	1
	b) Define java card. Also, explain with a neat diagram the components of Java Card Virtual Machine (JCVM) architecture.	08	L2	2	1
2.	a) Define Smart Card. Also, explain the benefits and applications of a smart card.	08	L2	1	1
	b) Demonstrate with a neat diagram the components of java card overall architecture.	08	L2	2	1
3.	a) Explain the essential characteristics of smart card communication model.	08	L2	1	1
	b) Demonstrate the significance of various java card applet file formats supported by java card deployment.	08	L2	2	1
Unit – II					
4.	a) Define transient object. Also, explain the necessity of APDU class in the java card APIs.	08	L1, L2	3	1
	b) Explain the approval process of the payment cards with a neat diagram.	08	L2	4	1

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|-------|--|----|----|---|---|
| 5. a) | Explain the transaction mechanism supported by java card technology. | 08 | L2 | 3 | 1 |
| b) | Illustrate EMV card / Chip architecture. | 08 | L2 | 4 | 1 |
| 6. a) | Summarize the essential properties of transient objects in java card technology. | 08 | L2 | 3 | 1 |
| b) | Explain the production and personalization of the payment cards with a neat diagram. | 08 | L2 | 4 | 1 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Explain along with INS about any four global platform (GP) commands. | 08 | L2 | 5 | 1 |
| b) | Illustrate with a neat diagram about card life cycle state transitions. | 08 | L2 | 5 | 1 |
| 8. a) | Illustrate along with INS about the necessity of content management and security management commands. | 08 | L2 | 5 | 1 |
| b) | Explain the process of secure card content management. | 08 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
